AP CALCULUS AB

Unit 1 • Self Check 1.1 and 1.2 • Lessons 1-1 to 1-4 • Answer key

Self Check 1.1 and 1.2 • Lessons 1-1 to 1-4

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$$\lim_{x \to -3} \frac{x^2 + x - 6}{x^2 - 9} =$$

(A) $\frac{5}{6}$

(B) 0

(C) $-\frac{5}{6}$

(D) It does not exist.

Solution

Answer: (A).

Question 2

NO CALCULATOR

$$\lim_{x \to \infty} \frac{2x - x^3}{4x^3 + 1} =$$

(A) 2

(B) $-\frac{1}{4}$

(C) 0

(D) It does not exist.

Solution

Answer: (B). Same degree: $\frac{-1}{4}$.

Question 3

NO CALCULATOR

$$\lim_{x \to -1} \frac{x^2 - 1}{|x + 1|}$$

(A) 2

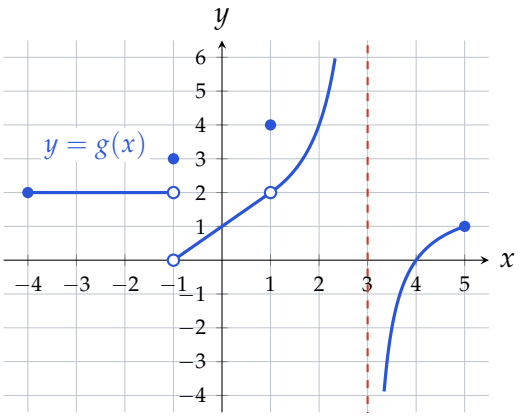
(B) -2

(C) 0

(D) does not exist

Solution

Answer: (D). Left 2, right -2.



Question 4

NO CALCULATOR

Using the graph of g, at which x is the discontinuity removable?

(A) $x = -1$
(B) $x = 3$
(C) $x = -4$
(D) $x = 1$

Solution
Answer: (D).

Question 5 (Free response)

NO CALCULATOR

Using the graph of g, classify the discontinuities at $x = -1$, 1 and 3, with limit statements.

Solution
-1: jump (limits 2 and 0). 1: removable ($\lim = 2 \neq 4 = g(1)$). 3: infinite ($+\infty$ and $-\infty$).